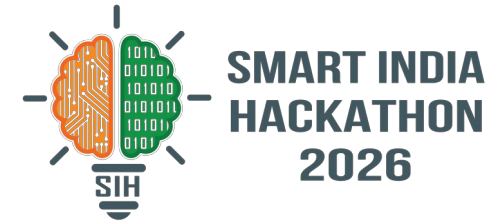


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26173
- **Problem Statement Title**- iTantra – Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**- 188881
- **Team Name (Registered on portal)** Team Turtle



Website



Demo film



Code



APK

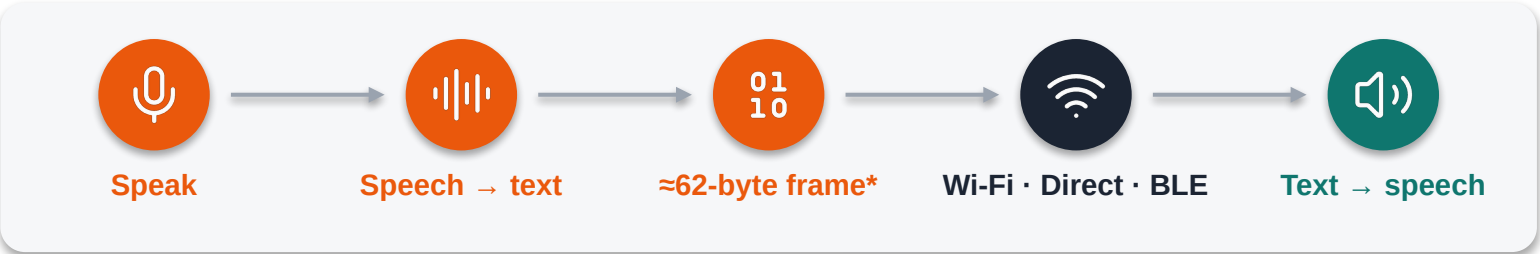


iTantra

Speak. Send text. Hear it back — offline.

Proposed Solution (Describe your Idea/Solution/Prototype)

Only text crosses the link; the receiver hears speech. **No tower. No internet. No cloud.**



Detailed explanation of the proposed solution

- 10 Indian languages, on the phone
- ALERT at alarm volume
- One-tap SOS with GPS/NavIC
- Push-to-talk and call mode

How it addresses the problem

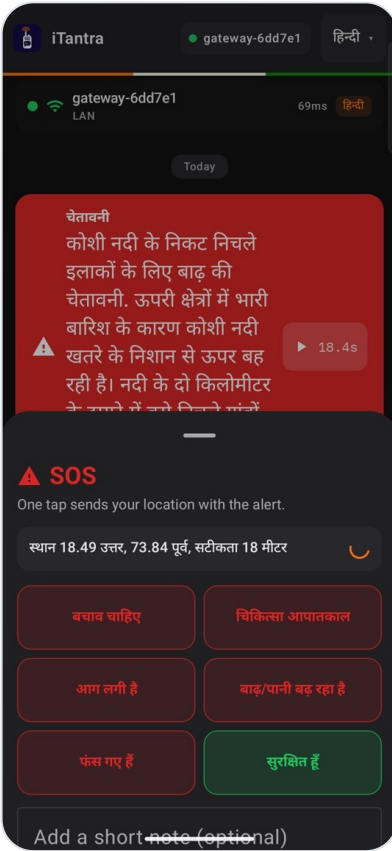
Format	Size (bits/s per sentence)
Raw audio	256k
Opus voice	12k
iTantra	165

bits/s per sentence (log scale)

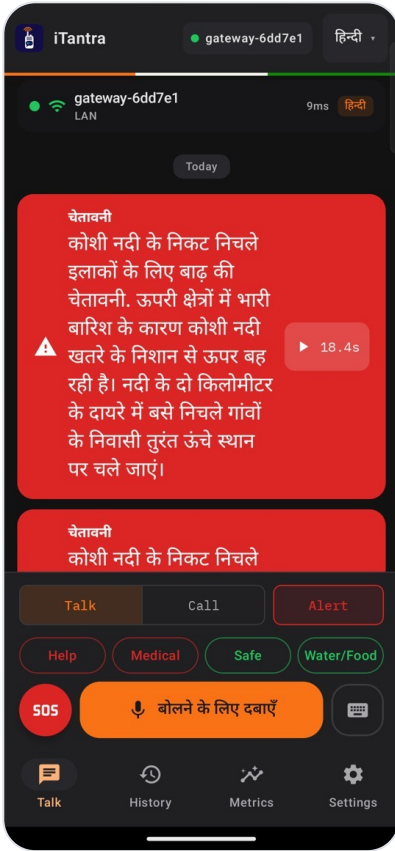
Thin links carry it. Output is still speech.

Innovation and uniqueness of the solution

- Speech itself is the codec
- Auto link fallback, one packet format
- Alerts cannot be interrupted
- LITE mode below 6 GB RAM



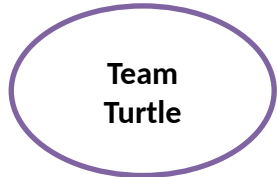
Sender · SOS



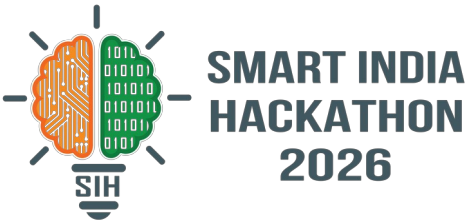
Receiver · ALERT

Watch the demo film · 1 min 55 s

**Median of 310 FLEURS sentences (unit test), before encryption and link headers. Bars assume a ~3 s sentence.*



TECHNICAL APPROACH

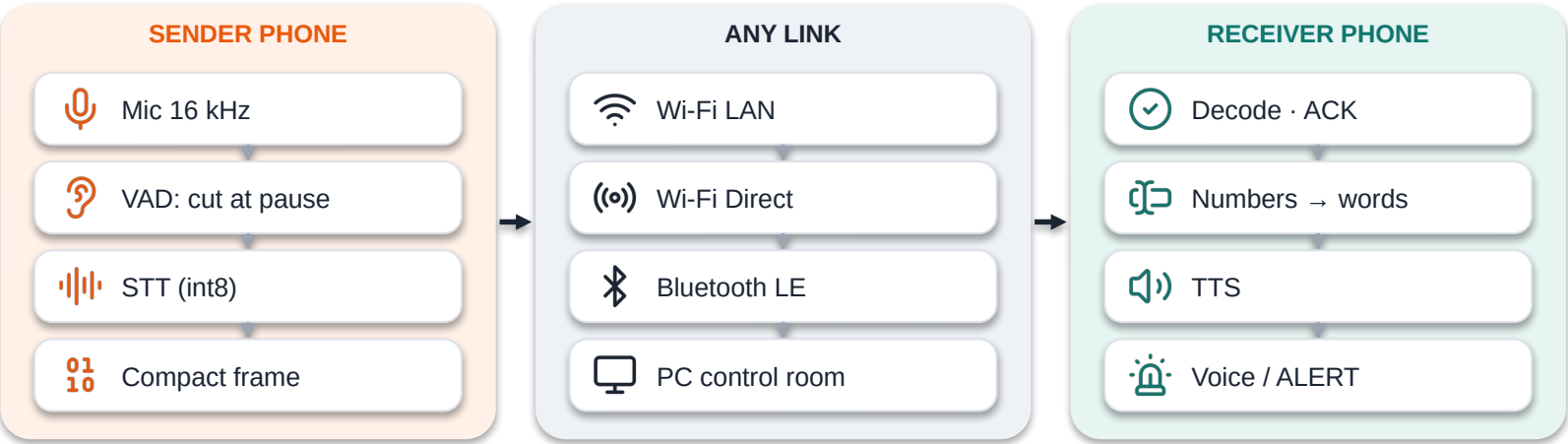


Technologies to be used (e.g. programming languages, frameworks, hardware)

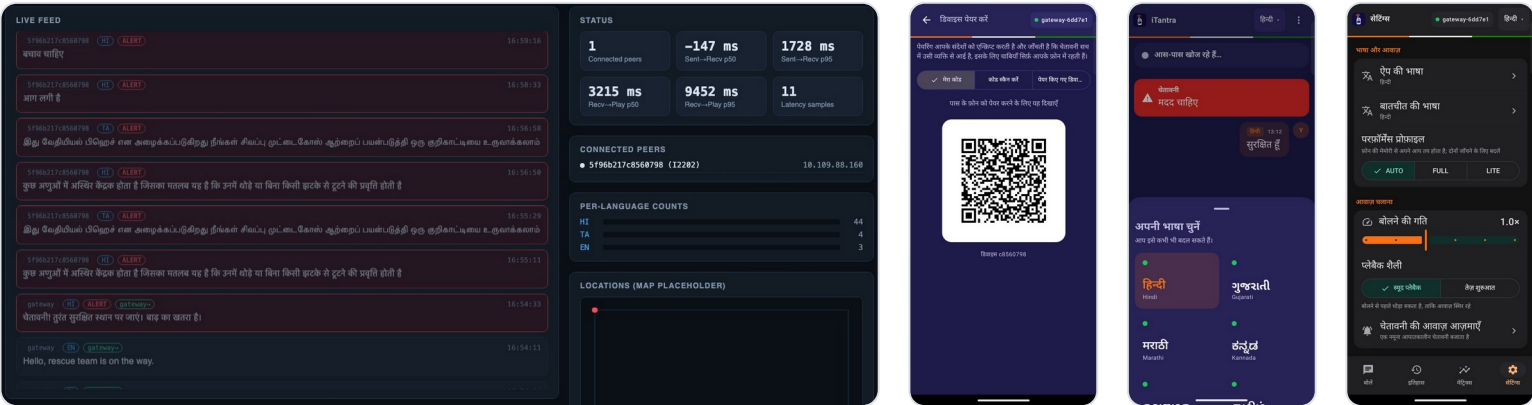
- Android app**
Kotlin · Jetpack Compose
- Voice detection**
Silero VAD
- Speech-to-text**
SraVaani-1.0 · sherpa-onnx
- Text-to-speech**
VITS Rasa · Indic-Mio
- Links**
Wi-Fi · Wi-Fi Direct · BLE
- Control room**
Python gateway · CAP 1.2

- BUILT** 10-language STT/TTS, ALERT, SOS
- TEST PENDING** Wi-Fi Direct, BLE phone-to-phone
- PLANNED** ESP32 LoRa bridge, smaller voices

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Working prototype — real screens

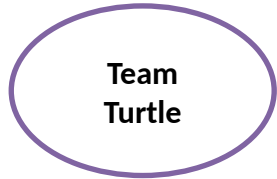


PC control room

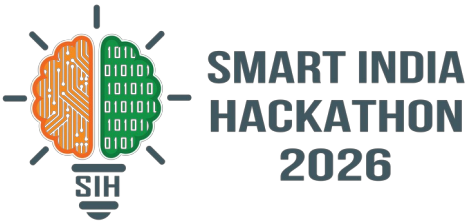
QR pairing

Language

LITE profile



FEASIBILITY AND VIABILITY



Analysis of the feasibility of the idea

Measured: 4 GB-class memory test, in-app logging



14×
faster than speech



0.6 s
speech end → text



765 MB
app RAM, idle



0.3%
idle CPU



Alive
app not killed

12 GB Snapdragon 870 phone, RAM locked to ~1.7 GB free, LITE profile, chip not slowed (a budget phone will be slower). Median of 10 languages; 1.8 GB peak while speaking; 115 other apps were closed by Android. WER 5.9% (preliminary, 10 clips).

Potential challenges and risks

Strategies for overcoming these challenges



Voice starts 3–6 s late in LITE



Keep listener's language loaded; smaller voices



1.14 GB of model files



Download per language on demand



Wi-Fi Direct / BLE not yet tested



Two-phone tests; Wi-Fi stays primary



Noise, accents, code-mixing



Alert-word boosting; typed-text fallback

Viability



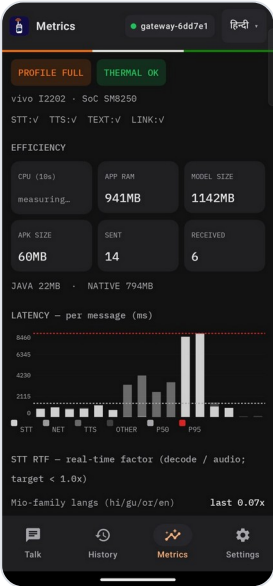
No airtime or cloud fees



Open models: MIT · Apache · CC-BY



Open code and measurements



In-app metrics (real)

Potential impact on the target audience



Disaster response teams

Towers and internet down



Coastal fishing communities

Weak links at sea



Low-literacy users

Hear it, don't read it



Remote field staff

Poor-coverage areas

≈1,500×

less data than raw audio*

1.03 s

ALERT arrival → spoken (Wi-Fi)

10

Indian languages, in and out

Benefits of the solution (social, economic, environmental, etc.)



Social

Alerts heard, not read



Economic

Phones people already own; no fees



Environmental

Fewer bytes on air

*62 B per ~3 s sentence (unit-test median) vs 256 kbps raw audio. No field pilot yet; battery gain not measured.

Photos, Wikimedia Commons: [NDRF – MHA, GODL-India](#) · [fishermen – Saral Shots, CC0](#) · [woman on phone – Ammalu, CC BY-SA 3.0](#) · [rail workers – David Brossard, CC BY-SA 2.0](#)

Details / Links of the reference and research work



Open models

[SraVaani-1.0 STT · ARTPARK, IISc](https://huggingface.co/ARTPARK-IISc/SraVaani-1.0)
huggingface.co/ARTPARK-IISc/SraVaani-1.0

[Indic-Mio TTS · SPRING Lab, IIT-M](https://huggingface.co/SPRINGLab/Indic-Mio)
huggingface.co/SPRINGLab/Indic-Mio

[VITS Rasa 13 TTS · AI4Bharat](https://huggingface.co/ai4bharat/vits_rasa_13)
huggingface.co/ai4bharat/vits_rasa_13

[Silero VAD](https://github.com/snakers4/silero-vad)
github.com/snakers4/silero-vad



Runtimes and libraries

[sherpa-onnx](https://github.com/k2-fsa/sherpa-onnx)
github.com/k2-fsa/sherpa-onnx

[llama.cpp / ggml](https://github.com/ggml-org/llama.cpp)
github.com/ggml-org/llama.cpp

[mio-tts-cpp](https://github.com/mmmnga/mio-tts-cpp)
github.com/mmmnga/mio-tts-cpp

[Bouncy Castle \(encryption\)](https://www.bouncycastle.org)
www.bouncycastle.org

[RadioLib \(LoRa\)](https://github.com/jgromes/RadioLib)
github.com/jgromes/RadioLib



Standards and data

[OASIS CAP 1.2 alerts](https://docs.oasis-open.org/emergency/cap/v1.2/CAP-v1.2.html)
docs.oasis-open.org/emergency/cap/v1.2/CAP-v1.2.html

[NDMA SACHET](https://sachet.ndma.gov.in)
sachet.ndma.gov.in

[Google FLEURS \(STT test set\)](https://huggingface.co/datasets/google/fleurs)
huggingface.co/datasets/google/fleurs

[Codec2 \(bitrate baseline\)](https://github.com/drowe67/codec2)
github.com/drowe67/codec2

[Meshtastic \(LoRa mesh\)](https://meshtastic.org)
meshtastic.org



[Website](#)
overview · results



[Demo film](#)
1 min 55 s



[Code](#)
GitHub



[APK](#)
Android 8+



Android guides

[Wi-Fi Direct](https://developer.android.com/develop/connectivity/wifi/wifi-direct) developer.android.com/develop/connectivity/wifi/wifi-direct

[Bluetooth and BLE](https://developer.android.com/develop/connectivity/bluetooth) developer.android.com/develop/connectivity/bluetooth

[Audio focus \(ALERT\)](https://developer.android.com/media/optimize/audio-focus) developer.android.com/media/optimize/audio-focus



Our work

[Code and measurements](https://github.com/prasadxa/itantra) github.com/prasadxa/itantra

[Website](https://itantra-106.pages.dev) itantra-106.pages.dev

[Demo film](https://itantra-106.pages.dev/demo) itantra-106.pages.dev/demo

[Model pack](https://r2.dev/models/v1/manifest.json) r2.dev/models/v1/manifest.json